

Attention-Aware Encoder–Decoder Neural Networks for Heterogeneous Graphs of Things

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I. INTRODUCTION

Abstract—Recent trend focuses on using heterogeneous graph of things (HGoT) to represent things and their relations in the Internet of Things, thereby facilitating the applying of advanced learning frameworks, i.e., deep learning (DL). Nevertheless, this is a challenging task since the existing DL models are hard to accurately express the complex semantics and attributes for those heterogeneous nodes and links in HGoT. To address this issue, we develop attention-aware encoder–decoder graph neural networks for HGoT, termed as HGAED. Specifically, we utilize the attention-based separate-and-merge method to improve the accuracy, and leverage the encoder–decoder architecture for implementation. In the heart of HGAED, the separate-and-merge processes can be encapsulated into encoding and decoding blocks. Then, blocks are stacked for constructing an encoder–decoder architecture to jointly and hierarchically fuse heterogeneous structures and contents of nodes. Extensive experiments on three real-world datasets demonstrate the superior performance of HGAED over state-of-the-art baselines.

Index Terms—Graph neural network (GNN), graph of things, heterogeneous graph, Internet of Things (IoT).

THE Internet of Things (IoT), where billions of various things are connected to the Internet, plays an important role in many domains, such as smart city, smart home, and smart manufacturing [1]–[4]. In IoT, things have diverse types as well as different relations to others things. For example, as shown in Fig. 1, there are multiple types of things, including people, vehicles, places, apps, and home appliances. The relations between things also have many types and rich semantic meanings. The relation between people and vehicles may be taking or driving, whereas between cellphones and apps is installing. These things and their relations naturally constitute the heterogeneous graph of things (HGoT) [2], [5] where things can be represented by nodes and relations are considered as edges. The relations of node pairs determine the edge types.

Deep learning (DL) has achieved great success both in dealing with the Euclidean data in domains, such as computer vision [6], [7] and natural language processing [8], [9], and in tackling the non-Euclidean graph data, such as graph convolutional network (GCN) [10], [11] and graph attention network (GAT) [12]. However, most existing deep models are inherently flawed in expressing the real-world HGoT since the contained nodes and links possess complicated and rich semantics [13].

Representation learning of HGoT remains an unmet challenge in current DL community. There are two most important properties need to be addressed as follows. First, heterogeneous nodes: Various types of nodes with different traits and features belong to different domains. The information in heterogeneous nodes is highly complicated and diverse. For example, the features of people may include sex, age, and nationality. Also, the vehicles may also have different types, such as car or bus. Second, heterogeneous semantics: Semantic information between heterogeneous nodes is highly complex with different influence on the traits of nodes [14]. Heterogeneous semantic is usually represented by metapaths [15], i.e., the metapath of people–vehicle–place has a meaning of travel, whereas the metapath of people–vehicle–people denotes a relationship of riding together. Therefore, how to fuse heterogeneous semantic information remains a big challenge [14].

In this article, we aim at solving the problems of node-level representation learning in HGoT. First, the features of nodes with different types are projected into a single feature space. Then, we propose a novel attention-based separate-and-merge method to handle the heterogeneous semantics. The rich and complicated semantics are processed separately first and then

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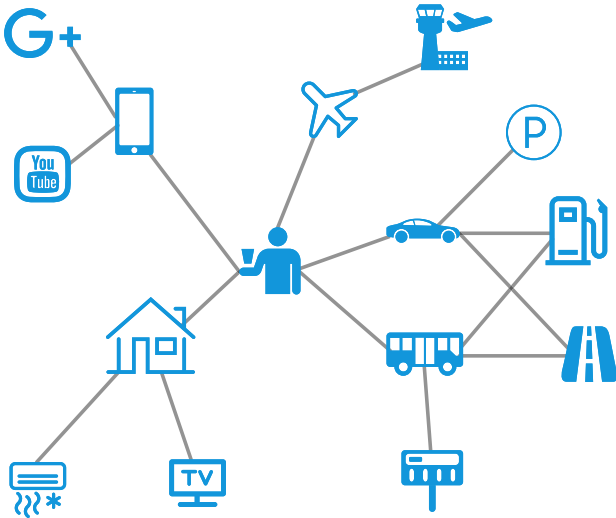


Fig. 1. Illustrative example of HGoT. There are various types of nodes, such as people, vehicles, places, etc. The relations between things also have various kinds and rich semantic meanings. These things and their relations naturally constitute the HGoT where things can be represented by nodes and relations are considered as edges.

merged together based on attention mechanism to generate new information. Specifically, the separate attention aims at aggregating the information within one type of semantics, whereas the merge attention targets on getting the optimal combination of different semantics. Through a single separate-and-merge attention process, the heterogeneity of nodes and rich semantic information can be fused jointly.

The encoder-decoder architecture, such as U-Net, has been successfully applied in many tasks because of its strong ability to capture and fuse the features hierarchically to enhance the representation learning ability [16], [17]. But it cannot deal with the heterogeneity and complexity of HGoT. Motivated by this, we attempt to design an encoder-decoder architecture for HGoT to learn the features of nodes and fuse the features hierarchically to enhance the representation learning ability. In the original encoder-decoder architecture, there are multiple encoding blocks and decoding blocks. Each encoding block consists of some convolution layers and pooling (down-sampling) layers, whereas each decoding block consists some convolution layers and up-sampling layers. There are also connections between the encoder and the decoder. However, due to the heterogeneity and complexity of HGoT, traditional processing of graph neural network (GNN) (e.g., graph convolution, pooling, and up-sampling) cannot be directly applied to HGoT.

The contributions of this article are listed as follows.

- 1) According to the attention-based separate-and-merge manner, we design a novel heterogeneous graph nodewise attention (hgaNode) layer, a heterogeneous graph attention pooling (hgaPooling) layer, a heterogeneous graph unpooling (hgUnpooling) layer, and unified connections (uConnections).
- 2) We develop a novel attention-aware encoder-decoder graph neural networks for HGoT, termed as “HGAED,”

that allows feature encoding, decoding, and fusion for representation learning of heterogeneous graphs.

- 3) Extensive experimental results on three real-world heterogeneous graph datasets demonstrate HGAED outperforms state-of-the-art baselines significantly.

The rest of this article is organized as follows. In Section II, related work is discussed. Section III describes the key definitions. Section IV presents the details of HGAED. Section V presents the experimental results. Finally, Section VI concludes this article.

II. RELATED WORK

The purpose of GNNs is to extend the deep neural networks to deal with arbitrary graph structured data [18] and has been used in many domains [19], [20]. GNNs are often viewed as a message passing algorithm that iteratively updates node representations by aggregating messages from neighboring nodes. In [10] and [11], GCN was designed based on spectral graph convolutions. All these works targeted on homogeneous graph. To the best of our knowledge, the latest framework graph U-Net [21] was the first encoder-decoder architecture for graphs. Nevertheless, it only learned from the homogeneous graphs and did not adopt the attention mechanism to take the heterogeneous node contents into consideration. Heterogeneous graph attention network (HAN) [22] proposed an attention-based neural network for heterogeneous graph; however, HAN does not incorporate the attention scheme and the encoder-decoder structure to capture the rich semantic features in HGoT.

There are some unsupervised methods to learn the node representation in the literature. DeepWalk [23] samples the training pairs from random walk and learns the node embeddings in a SkipGram way for homogeneous graphs. For heterogeneous graph embedding, metapath2vec [24] uses a biased random walk scheme to incorporate the semantic information. SHNE [25] proposed a heterogeneous SkipGram model to store both the structural features and multiple types of nodes and relations in the embedding space. The major difference between these methods and GNN-based methods, including our HGAED, is that they learn embedding for each node while the parameters in GNNs are shared for all the nodes and can be generalized to unseen nodes and graphs.

Pooling layers can enable CNN models to scale down the size of representations and, thus, avoid overfitting and learn the coarse features. DiffPool [26] was the first to propose a learned pooling layer for graph data. Gao and Ji[21] proposed graph U-Net that utilized top-k nodes for the graph. Lee *et al.* [27] introduced self-attention graph pooling, a variant of top-k pool based on attention. The previous work for graph pooling in the literature targeted on homogeneous graphs only.

Attention mechanisms [28] have become one of the most influential mechanisms in DL. In [12], GAT was proposed that achieved state-of-the-art performance on several homogeneous graph node classification tasks. Wang *et al.* [22] extended GAT for presentation learning of heterogeneous graphs. However, it was not an encoder-decoder architecture and there were no pooling operations that can coarsen features.

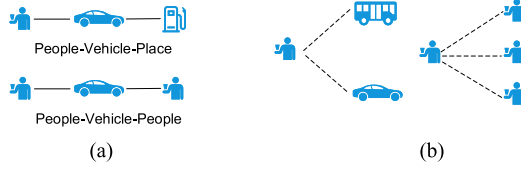


Fig. 2. Illustration of metapath and metapath-based neighbors. (a) Metapath. (b) Metapath-based neighbors.

III. PRELIMINARY

In this section, we list some key definitions for HGoT in the following.

Definition 1: HGoT: A HGoT is defined as $G = (V, E)$, where E is the set of edges and V denotes the set of nodes. A is the set of object types and R is the set of relation types. For a HGoT, $|A| > 1$ and $|R| = 1$.

Definition 2: Metapath: A metapath P_i in the metapath set P of the graph G is defined as $A_1 \rightarrow A_2 \rightarrow \dots \rightarrow A_{l+1}$, or $A_1 A_2 \dots A_{l+1}$ for short, which defines a composite relation $R = R_1 \circ R_2 \circ \dots \circ R_l$ between objects A_1 to A_{l+1} , where $\circ$ denotes the composition operator on relations. In HGoT, two nodes can be connected via different metapaths, which are utilized to denote the semantic connections between nodes.

Fig. 2(a) illustrates two metapaths, the metapath of people-vehicle-place has a meaning of travel, whereas the metapath of people-vehicle-people denotes a relationship of riding together.

Definition 3: Metapath-based neighbors: Given a node i and a metapath P_k in a HGoT, where k is the index of the metapath. The metapath-based neighbors $N_i^{P_k}$ of node i are defined as the set of nodes that connect with node i via metapath P_k .

Shown in Fig. 2(b), if people-vehicle and people-people are defined metapaths, then the connected vehicles, including bus and car, as well as people, are the metapath-based neighbors of people.

Definition 4: Metapath-based subgraph: Given a metapath P_k in a HGoT, the metapath-based subgraph is defined as $SG^{P_k} = (V, E^{P_k})$, where V is the set of nodes in a HGoT and E^{P_k} is the set of edges under metapath P_k , which is defined as a metapath-based adjacent matrix M^{P_k} .

We can construct a metapath-based subgraph for each metapath. Each metapath-based subgraph can reflect one type of semantics.

IV. PROPOSED METHODOLOGY

The architecture of HGAED is illustrated in Fig. 3. HGAED takes the unified features of different types of nodes, metapath-based subgraphs as inputs. HGAED is based on our designed hgaNode, hgaPooling, hgUnpooling, and uConnections. There are an encoder part, a decoder part, and uConnections between them.

Several encoding blocks build the encoder part. Each of blocks consists of a hgaNode layer followed with a hgaPooling layer. The hgaNode layer updates each node's features through aggregating the heterogeneous features from its neighboring nodes under different semantics by the separate-and-merge attention.

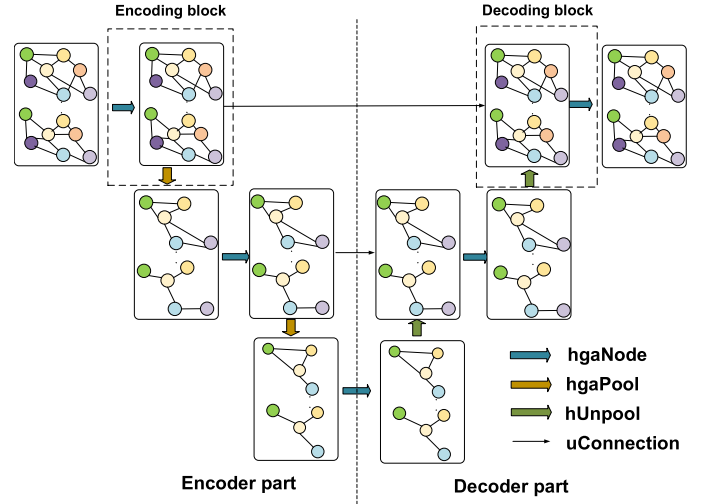


Fig. 3. Architecture of the proposed HGAED. It is an example of HGAED with two blocks in encoder and decoder.

In hgaPooling layer, we first utilize the separate-and-merge attention to get the self-attention score for each node that estimates the information that a node can retain. Then, we select the nodes that can retain more information in the HGoT. Through the hgaPooling layer, the sizes of the metapath-based subgraphs are reduced and the higher level features of the selected node are summarized. Several encoding blocks are stacked to form the encoder part, which down-samples and learns the features of the HGoT, and fuse heterogeneous semantics hierarchically.

The number of decoding blocks is the same as that in the encoder. Each decoding block has a hgaNode layer followed with a hgUnpooling layer. As an inverse operation of the hgaPooling layer, the hgUnpooling layer restores the HGoT to its original structure with the help of locations of nodes selected in the corresponding hgaPooling layer.

There are uConnections used to connect encoding blocks and corresponding decoding blocks, which can fuse the coarse-grained unified features in the encoders and the fine-grained unified features in the decoders. In our implementation, we combine the node features of encoding blocks and that of corresponding decoding blocks by addition.

A. Representation of Heterogeneous Nodes

The feature space of different types of nodes varies. Therefore, for each node type, we utilize the type-specific projection method to map the features of different node types into the same feature space

$$h_j = W_{A_i} \cdot h'_j \quad (1)$$

where W_{A_i} is the transformation matrix for node type A_i , and h'_j and h_j are the original and the projected features of node j , which is type A_i .

B. Design of hgaNode

In our proposed hgaNode, features of each node update recursively, to capture the features of its neighboring nodes and

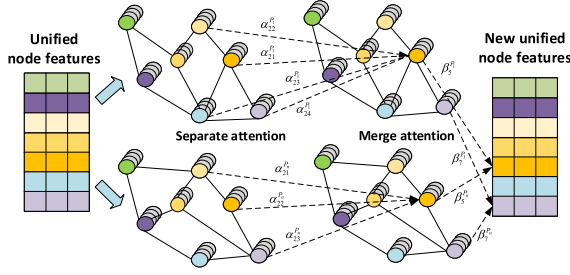


Fig. 4. Illustration of hgaNode.

the heterogeneous network structure. The process of hgaNode is shown in Fig. 4. The hgaNode takes the unified features of all the nodes and metapath-based subgraphs as inputs and contains two attention processes: separate attention and merge attention. The purpose of separate attention is to learn the weights of metapath-based neighbors and aggregate them to form the new node features for each metapath-based subgraph. After that, hgaNode can distinguish the difference of multiple semantics via merge attention. Based on the two-level attention, hgaNode can get the optimal combination of neighbors in multiple metapaths, which enables the learned node features better capturing the complicated structure and rich semantic information in a HGoT.

1) Separate Attention: That targets on aggregating the features of metapath-based neighbors for each node under a specific kind of metapath to form semantic-specific node embedding. It is semantic-specific and aims to capture one kind of semantic information. Given a node pair (i, j) connected via metapath P_k , the separate attention $e_{ij}^{P_k}$ is to learn the attention coefficients that means the importance of node j to node i under metapath P_k

$$e_{ij}^{P_k} = \sigma(W^{P_k} h_i, W^{P_k} h_j) \quad (2)$$

where $e_{ij}^{P_k}$ indicates the importance of node j 's features to node i under the metapath P_k , W^{P_k} is the shared weighted matrix of the metapath P_k , and σ is the activation function (e.g., *relu*). We inject the metapath-based subgraph structure of the metapath P_k into the mechanism by performing masked attention: we only compute e_{ij} for node i 's metapath P_k based neighbors.

Then, the semantic-specific embedding of node i can be aggregated by the metapath-based neighbor's projected features with the corresponding coefficients. Following the standard settings of attention network, we use the rowwise softmax operator to normalize them, and obtain the final weight coefficients $\sigma_{ij}^{P_k}$ as follows:

$$\alpha_{ij}^{P_k} = \text{softmax}(e_{ij}^{P_k}) = \frac{\exp(e_{ij}^{P_k})}{\sum_{l \in N_i^{P_k}} \exp(e_{il})} \quad (3)$$

where $N_i^{P_k}$ is metapath-based neighbors of node i under metapath P_k . The metapath-based embedding of node i under the metapath P_k can be aggregated by the neighbor's projected features with the corresponding coefficients as follows:

$$\tilde{h}_i^{P_k} = \sigma\left(\sum_{j=1}^{N^{P_k}} \alpha_{ij} h_j\right) \quad (4)$$

where $\tilde{h}_i^{P_k}$ denotes the obtained semantic-specific features of node i under metapath P_k , and N^{P_k} is the total of nodes in metapath-based graph $SG_k^{P_k}$. Taking $|P|$ types of metapaths as an example, after the separate nodewise attention, $|P|$ groups of semantic-specific node embedding $\{\tilde{h}^{P_1}, \tilde{h}^{P_2}, \dots, \tilde{h}^{P_{|P|}}\}$ can be obtained.

2) Merge Attention: Commonly, each node in a HGoT incorporates many kinds of semantic information. However, semantic-specific node embedding can only partly represent a node. Different metapaths contain multiple semantics and we need to merge them to learn a more comprehensive node embedding [13]. We develop a novel merge attention scheme that can automatically learn and fuse the importance of different semantics.

In separate attention, we consider each pair of node that are connected by a specific metapath. However, in merge attention, we explore the dependence between each semantic-specific node embedding and the entire q groups of node embedding, and merge these groups of node embedding into a unified node embedding space. The importance of each metapath e^{P_k} is calculated as follows:

$$e^{P_k} = o^T \sigma(W \tilde{h}^{P_k} + b) \quad (5)$$

where b is the bias vector, W is the weight matrix, and o is the semantic-level attention vector. All the metapaths share the aforementioned parameters.

Similar with separate nodewise attention, the importance of all metapaths are normalized with softmax function and obtain the final weight coefficients β^{P_k} for each metapath. Apparently, the higher β^{P_k} , the more important metapath P_k is. These semantic-specific embeddings are then fused to form the final embedding h_{out}^0 with the learned weights, which is the output of hgaNode, as follows:

$$h_{\text{out}}^0 = \sigma\left(\sum_{k=1}^{|P|} \beta^{P_k} \tilde{h}^{P_k}\right). \quad (6)$$

C. Design of hgaPooling

For the grid-like data in CNNs, pooling layers are of great importance. The sizes of feature maps can be reduced and the receptive fields can be enlarged, which will result in better generalization and performance. Similarly, the pooling operation in graph is to select and maintain part of the node features for the next layer to process. Gao and Ji employed gPool [21] and have achieved a comparable performance to DiffPool [26] for graph. However, gPool was designed only for homogeneous graph, and the graph topology was not considered during the pooling operation. We propose hgaPooling for HGoT to address these issues. As shown in Fig. 5, the inputs of hgaPooling are the unified features of all the nodes and the metapath-based subgraphs. The outputs of hgaPooling are the pooled unified features of the selected nodes and the reduced metapath-based subgraphs. There are two subprocesses in hgaPooling, structure-aware attentional projection and attentional heterogeneous graph pooling, as discussed in the following.

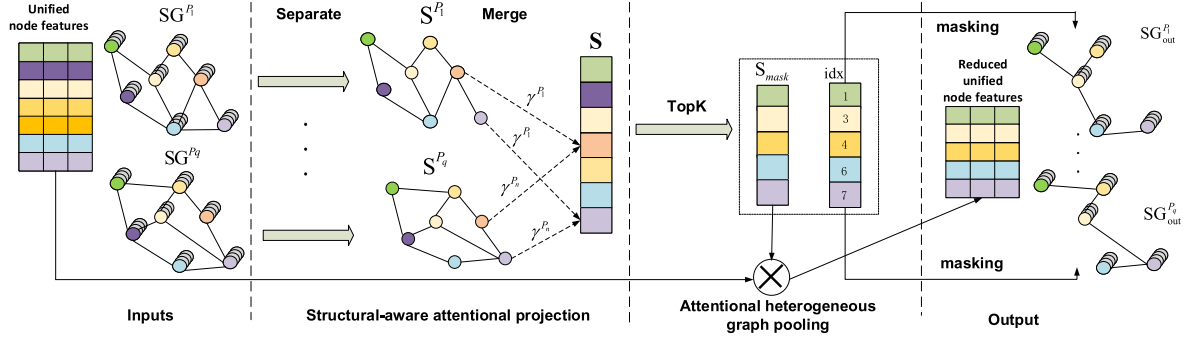


Fig. 5. Illustration of hgaPooling.

1) **Structure-Aware Attentional Projection:** During hgaPooling, we first utilize the separate-and-merge attention mechanism and also consider the topology of the HGoT to get the self-attention score for each node, which measures the information of a node retaining when projected into a learnable direction vector.

First, we obtain the semantic-specific self-attention scores using graph convolution for each metapath-based subgraph by separate attention. Through this operation, the semantic-specific self-attention scores are based on both node features and topology of metapath-based subgraph. Following graph convolution formula of Kipf and Welling [11], the semantic-specific self-attention score $\tilde{s}^{P_k}$ is calculated as follows:

$$\tilde{s}^{P_k} = \sigma \left(\tilde{D}^{P_k - \frac{1}{2}} \tilde{M}^{P_k} \tilde{D}^{P_k - \frac{1}{2}} h W^1 \right) \quad (7)$$

where h are the unified node features, $\tilde{M}^{P_k}$ is the adjacency matrix of the semantic-specific subgraph SG^{P_k} with added self-loop, $\tilde{D}^{P_k}$ is the degree matrix of $\tilde{M}^{P_k}$, and $W^1 \in \mathbb{R}^{F \times 1}$ is a trainable vector that denotes a linear transformation to change the dimensionality of the features to scores.

After the separate projection for each metapath-based subgraph, q groups of semantic-specific self-attention scores $\{\tilde{s}^{P_1}, \tilde{s}^{P_2}, \dots, \tilde{s}^{P_q}\}$ can be obtained. Then, we explore the dependence between semantic-specific self-attention node scores and the entire q groups of node scores, and merge the q groups of node scores into a single node score. The importance of each metapath for the self-attention node scores, denoted as a^{P_k} , is calculated using the way in (5).

The weight of metapath P_k for the node scores, denoted as γ^{P_k} , can also be obtained by normalization and softmax function, as described in (11). With the learned weights as coefficients, we can fuse these semantic-specific self-attention scores to obtain the final score s

$$s = \sigma \left(\sum_{k=1}^{|P|} \gamma^{P_k} \tilde{s}^{P_k} \right). \quad (8)$$

2) **Attentional Heterogeneous Graph Pooling:** We use the node selection method proposed by Gao and Ji [21], where a portion of nodes is retained in the input of the HGoT. The number of nodes to be remained is determined by the pooling ratio $k \in (0, 1]$. The top $\lceil kN \rceil$ nodes are chosen by the value of

s . First, we obtain the indices of the selected nodes and then get the self-attention scores of the nodes based on the indices

$$idx = \text{topk}(s, \lceil kN \rceil) \quad (9)$$

where the topk function returns the indices of the top $\lceil kN \rceil$ nodes.

After obtaining the self-attention scores and the indices of the nodes that are required to be retained, the output node features of the selected nodes and the reduced adjacent matrices can be calculated as follows:

$$h_{out}^1 = h_{out,idx}^0 \odot s, M_{out}^{P_k} = M_{idx,idx}^{P_k} \quad (10)$$

where $h_{out,idx}^0$ is the nodewise indexed feature matrix, which retains the nodes with idx indices, h_{out}^1 is the output feature matrix, $\odot$ is the broadcasted elementwise product, and $M_{idx,idx}^{P_k}$ is the rowwise and colwise indexed adjacency matrix, whereas $M_{out}^{P_k}$ is the reduced adjacent matrix of metapath P_k .

D. Design of hgUnpooling

We propose the hgUnpool layer, which is an up-sampling operation, performing the inverse operation of the hgaPooling layer to restore the original structure. The idx are recorded, which denotes the node locations chosen in the corresponding hgaPooling layer. The aim of idx to restore nodes to the original positions in the graph, as shown in the following:

$$h_{out} = \text{Distribute}(0_N, h_{out}^1, idx) \quad (11)$$

where h is the input of hgUnpooling, which is the original node feature matrix, h_{out} is the output of hgUnpooling, 0_N is initially empty feature matrix of the new up-sampled graph, N is the number of nodes of the new up-sampled graph, and Distribute denotes the function that can distribute node features in h to 0_N based on the idx .

Through this operation, nodes with indices of idx are updated by node features in h , whereas the features of other nodes remain zero. We also record the edges reduced in the corresponding hgaPooling layer and put the edges back for each metapath-based subgraph. One thing to note is that metapath-based attention weight is not need in the unpooling operation because the semantic-specific features have been captured in the hgaNode and hgaPooling layers.

TABLE I
DESCRIPTION OF THE DATASETS

Dataset	Relations (A-B)	Number of A	Number of B	Number of A-B	Feature	Training	Validation	Test	Meta-paths
ACM	Paper-Author	3025	5835	9744	1830	600	300	2125	PAP
	Paper-Subject	3025	56	3025					PSP
DBLP	Paper-Author	14328	4057	19645	334	800	400	2857	APA
	Paper-Conf	14328	20	14328					APCPA
	Paper-Term	14327	8789	88420					APTPA
IMDB	Movie-Actor	4780	5841	14340	1232	300	300	2687	MAM
	Movie-Director	4780	2269	4780					MDM

E. Loss Function

Our proposed HGAED is end-to-end trainable. We adopt the node classification task to supervise the training process and the negative log-likelihood function as the objective function, which is formulated as follows:

$$L = \sum_{v \in V} y_v \log(\sigma(h_{\text{out}}^v)) + (1 - y_v) \log(1 - \sigma(h_{\text{out}}^v)) \quad (12)$$

where v is a node in graph node set V , y_v denotes the label of v , h_{out}^v is the feature of v after unpooling, and σ is the sigmoid activation.

V. EXPERIMENTAL RESULTS

A. Datasets

The datasets are described as follows.

- 1) BLP:¹ A subset of DBLP. We categorize the authors as the following four areas: database, data mining, machine learning, and information retrieval.
- 2) ACM:² We extract papers published in KDD, SIGMOD, SIGCOMM, MobiCOMM, and VLDB, and we categorize the papers as the following three classes: database, wireless communication, and data mining.
- 3) IMD: A subset of IMDB and the movies are categorized as three classes: action, comedy, and drama, according to their genre.

Table I illustrates the details. The terms “APA,” “PAP,” “MAM,” etc., denote different kinds of metapaths where “A” refers to author, “P” refers to paper, etc. HGAED can be easily generalized to other HGoT datasets by changing the metapaths in different scenarios, for example, using people-vehicle-place instead of author-place-author.

B. Baselines

We compare HGAED with some state-of-the-art baselines described as follows.

- 1) Network embedding methods for homogeneous graphs: DeepWalk [23]. The node heterogeneity is ignored and we perform DeepWalk on the whole heterogeneous graph.
- 2) Network embedding methods for heterogeneous graphs: ESIm [29], metapath2vec [24], and HERec [30]. The metapaths for metapath2vec and HERec are tested, and then we report the best performance.
- 3) GNN-based methods for homogeneous graphs include GCN [11], GAT [12], and graph U-Net [21]. For GCN,

¹[Online]. Available: <https://dblp.uni-trier.de>

²[Online]. Available: <http://dl.acm.org/>

TABLE II
EXPERIMENTAL RESULTS OF THE CLASSIFICATION TASK

Methods	ACM		DBLP		IMDB	
	MaF1	MiF1	MaF1	MiF1	MaF1	MiF1
DeepWalk	84.17	83.88	84.81	86.26	50.35	54.33
ESim	83.00	82.89	92.53	93.44	32.06	35.59
metapath2vec	73.81	73.69	91.89	92.80	45.15	48.81
HERec	73.92	73.84	92.34	93.27	47.64	50.99
GCN	88.29	88.35	92.38	93.09	51.81	54.61
GAT	87.33	87.11	91.73	92.55	52.99	56.97
HAN	89.04	88.80	92.30	93.04	54.38	58.51
Graph U-Net	88.20	88.05	92.31	93.20	53.49	58.43
HGAED	90.33	90.09	93.46	94.09	55.48	59.26

GAT, and graph U-Net, we test all the metapaths and report the best performance.

- 4) GNN-based methods for heterogeneous graphs: HAN [22].

C. Implementation Details

HGAED is implemented on PyTorch.¹ In HGAED, the same parameter setting as the source codes released in GAT and HAN is adopted. Specifically, we set the optimization model as Adam, the learning rate to 0.005, and the regularization parameter to 0.001. We adopt multihead attention and the number of attention head K to 8. This setting is also utilized in GAT and HAN for fair comparison. The network depth is two layers of encoding and decoding blocks, which is the same as GAT, HAN, and graph U-Net. The pooling percentage in HGAED and graph U-Net is set to 60%. For GNN, the training, validation, and test set are split the same to ensure fairness. The training ends if the validation loss does not decrease after 100 consecutive epochs. For random walk based models, including DeepWalk, ESIm, metapath2vec, and HERec. We adopted the results in [22].

D. Main Results

- 1) **Classification:** The KNN classifier with $k = 5$ is adopted to perform node classification for all the node embeddings obtained by the algorithms, which is utilized in HAN, to ensure fairness. The experiments are repeated for ten times and we report the averaged Macro-F1 (Ma-F1) and Micro-F1 (Mi-F1), which are also the evaluation metrics utilized in HAN. The classification results are shown in Table II.

For heterogeneous graph embedding methods, ESIm that can utilize multiple metapaths performs better than metapath2vec and HERec. GNN-based methods that incorporates both the node feature and the structure proved to be better than graph

¹The source code as well as the dataset will be provided upon request. It will be released to the public after this article review.

TABLE III
EXPERIMENTAL RESULTS OF THE CLUSTERING TASK

Methods	ACM		DBLP		IMDB	
	NMI	ARI	NMI	ARI	NMI	ARI
DeepWalk	41.61	35.10	76.53	81.35	1.45	2.15
ESim	39.14	34.32	66.32	68.31	0.55	0.10
metapath2vec	21.22	21.00	74.30	78.50	1.20	1.70
HERec	40.70	37.13	76.73	80.98	1.20	1.65
GCN	51.40	53.01	75.01	80.49	5.45	4.40
GAT	57.29	52.63	71.50	77.26	8.45	7.46
HAN	53.56	52.58	72.12	76.64	10.87	10.01
Graph U-Net	51.86	50.85	75.52	80.70	10.46	10.02
HGAED	58.46	59.09	78.84	84.62	10.96	10.41

TABLE IV
ABLATION STUDIES OF THE HGAED

Methods	ACM		DBLP		IMDB	
	MaF1	MiF1	MaF1	MiF1	MaF1	MiF1
NoAtt	88.32	88.41	92.38	92.97	51.78	54.21
SeparateAtt	88.82	88.66	92.01	93.13	52.42	55.69
MergeAtt	88.79	88.51	92.78	93.21	52.55	57.43
HGAED	90.33	90.09	93.46	94.09	55.48	59.26

embedding methods. Graph U-Net performs better than GCN due to the adoption of encoder–decoder architecture. HAN performs better than GAT that demonstrates the effectiveness of considering heterogeneous semantics.

We observe that our proposed HGAED achieves the best performance on all the datasets. The results demonstrate that the combination of separate-and-merge attention mechanism and encoder–decoder architecture can improve the representation learning for HGoT.

2) Clustering: We use the clustering task to assess the quality of the learned embeddings of the aforementioned models. We use the K -means algorithm for node clustering. The number of clusters K is equal to the number of classes. NMI and ARI are adopted to evaluate the quality of the clustering results. The experiments are repeated for ten times and we report the averaged results in Table III. We can find that HGAED performs consistently much better than other methods on all the datasets.

E. Ablation Studies

In this section, some ablation studies on node classification are explored to evaluate the effectiveness of the proposed separate-and-merge attention and the adoption of encoder–decoder architecture for HGoT.

1) Ablation Studies of the hgaNode: We evaluate the following variants regarding to hgaNode.

- 1) NoAtt means utilizing GCN for each metapath-based subgraph and then merge them by the average manner.
- 2) SeparateAtt means only utilizing the attention mechanism to aggregate the node features of metapath-based neighbors for each metapath separately and then aggregating the new node embeddings of each metapath in average.
- 3) MergeAtt stands for utilizing GCN for each metapath-based subgraph and then merging them by the attention mechanism.

The results are presented in Table IV. We can observe that if the attention mechanism is not adopted for the nodewise aggregation, the performance is decreased significantly. SeparateAtt

TABLE V
ABLATION STUDIES OF THE HGAPOOLING AND HGUNPOOLING

Methods	ACM		DBLP		IMDB	
	MaF1	MiF1	MaF1	MiF1	MaF1	MiF1
NOPool	81.87	81.41	89.13	81.21	51.81	54.61
GPool	87.40	87.19	92.63	91.72	53.17	53.57
SeparatePool	89.20	89.04	93.13	93.92	54.36	53.79
HGAED	90.33	90.09	93.46	94.09	55.48	59.26

TABLE VI
ABLATION STUDIES OF THE UCONNECTIONS

Methods	ACM		DBLP		IMDB	
	MaF1	MiF1	MaF1	MiF1	MaF1	MiF1
NOCon	87.90	89.60	85.98	86.84	54.47	56.79
HGAED	90.33	90.09	93.46	94.09	55.48	59.26

and MergeAtt perform better than NoAtt, which demonstrates the effectiveness of adopting the separate attention and merge attention, respectively. HGAED achieves the best results on all datasets. It demonstrates that it is quite important to capture both the importance of nodes and that of semantics in the HGoT analysis.

2) Ablation Studies of hgaPooling and hgUnpooling: We evaluate the following variants regarding to hgaPooling.

- 1) NOPool means that we do not utilize the hgaPooling in HGAED.
- 2) GPool denotes utilizing the projection method proposed in [21] to get node scores of all the nodes, then utilize $top-k$ method for pooling.
- 3) SeparatePool stands for utilizing our proposed structure-aware projection to obtain the node scores for each metapath-based subgraph and merge them in average.

From the results, as shown in Table V, we can observe that if hgaPooling is not adopted, the performance is decreased significantly. These results demonstrate the contributions of hgaPooling and hUnpooling layers for performance improvement. The proposed hgaPooling and hUnpooling encode higher level features, therefore achieve better generalization and performance. HGAED performs better than GPool and SeparatePool. That is because we can obtain more meaningful scores of the nodes based on the separate-and-merge attention in hgaPooling.

3) Ablation Studies of uConnections: We evaluate the variant NOCon that means we do not adopt the uConnections in the architecture. The results are presented in Table VI, which shows that the adoption of uConnection can improve the performance significantly. It demonstrates the effectiveness of fusing coarse-grained and fine-grained features by uConnections.

F. Experiments on Parameters

1) Network Depth: We conduct experiments to study the relationship between performance and network depth in terms of node classification accuracy. The results are reported in Table VII. We observe that different datasets are appropriate for adopting different depths. Specifically, HGAED with two blocks achieves the best results on ACM and IMDB, whereas HGAED with four blocks performs best on DBLP. That is due to the fact that DBLP is more complex than ACM and IMDB.

TABLE VII
COMPARISON OF DIFFERENT NETWORK DEPTHS

Depths	ACM		DBLP		IMDB	
	MaF1	MiF1	MaF1	MiF1	MaF1	MiF1
2	90.33	90.09	93.46	94.09	55.48	59.26
3	84.72	84.56	93.43	94.14	55.25	59.14
4	86.49	86.54	94.37	94.20	54.99	58.77
5	85.89	86.26	94.16	94.03	54.76	58.54

TABLE VIII
COMPARISON OF DIFFERENT NUMBER OF NODES
RETAINED IN THE POOLING

Percentage	ACM		DBLP		IMDB	
	MaF1	MiF1	MaF1	MiF1	MaF1	MiF1
40%	89.97	89.84	93.28	93.29	52.15	56.46
60%	90.33	90.09	93.46	94.09	55.48	59.26
80%	90.49	90.01	93.48	94.01	55.38	59.19

2) *Number of Nodes Retained in the Pooling*: In these experiments, we evaluate the performance of HGAED with the varied numbers of nodes retained in hgaPooling. The results of node classification accuracy are summarized in Table VIII. We can observe that retaining 60% of the nodes performs a little better than 40%. The results of retaining 80% of the nodes are similar with that of 60%.

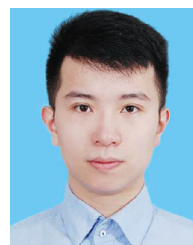
VI. CONCLUSION

In this article, we targeted the representation learning of HGoT. We proposed to utilize attention-based separate-and-merge manner to address the heterogeneity of nodes and semantics. Based on this idea, we designed the operations of hgaNode layer, hgaPooling layer, hgUnpooling layer, and hConnection. Multiple layers were stacked to construct an encoder-decoder architecture, which can aggregate the features of various types of nodes with rich semantics hierarchically. The proposed HGAED can capture the complex structures and rich semantics of heterogeneous graphs. Experimental results on node classification and clustering tasks demonstrated the effectiveness of HGAED.

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